Charting the Future of Global Health

Welcome to the IGHS AI Lab!

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September 1, 2026

UCSF Institute for Global Health Sciences

An AI Lab (No Lab Coat Required)

- Solve real problems
- Discover something new
- Explore out loud
- Lead with curiosity
- Keep it fun
- https://tinyurl.com/ighs-ai-lab

Today's Agenda

- Article discussion: "Safety of a large language model-based clinical decision support system in African primary healthcare"
  - Alexander Marr, PhD, MPH, Center for Applied Public Health
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC13246507/
- Q&A

Upcoming

Date	Topic	Discussant(s)
9/15/2026	Considering AI-enabled research and funding platforms	Gabe Murphy & Vincent Chen (OCR) and IGHS
9/29/2026	ChatGPT and MS Office plugins	Aaron McCoy, IGHS
10/13/2026	Using AI for QUAL research, opportunities & potential pitfalls	James Goh, Evidano
10/27/2026	Looking for Discussion Leads	TBD

Upcoming AI Office Hours at IGHS

- Bi-weekly Tuesdays, 12-1pmPT
- Next is Tuesday, September 15
- Let me know if you want to present or lead a discussion topic
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IGHS at 25

Charting the Future of Global Health

Article Discussion

Alex Marr

IGHS CAPH

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IGHS AI Lab

Safety of a Large Language Model-Based Clinical Decision Support System in African Primary Healthcare

Agweyu, Mwaniki, Musau, Korom, et al. • Nature Health, Vol 1, June 2026
Presented by: Alexander Marr

A retrospective safety evaluation of an EMR-embedded GPT-4o clinical decision support tool (“AI Consult”), deployed across 16 Penda Health primary care clinics in Nairobi and Kiambu, Kenya.

AI statement: Claude Sonnet 5.0 was used to format this presentation and generate the figures.

The Problem This Study Responds To

1

Workforce gaps

Sub-Saharan Africa faces the world's most acute health worker shortages, with limited access to specialist mentorship for frontline clinicians.

2

LLMs as a potential bridge

LLM-based clinical decision support (CDSS) could deliver real-time, specialist-grade diagnostic and management suggestions to under-resourced clinics.

3

But evidence is thin — and risky

Most LLMs are trained on high-income-country data. 'Naive' deployment risks flawed guidance (wrong drugs, unavailable tests) that clinicians may not catch.

Prior evidence gap: no published clinical evaluation of a real-world, EMR-embedded LLM CDSS — in any setting — existed before this study.

The Setting: Penda Health & the “AI Consult” Tool

Study Setting

16 outpatient clinics (Penda Health), Nairobi & Kiambu, Kenya

78,366 clinical consultations in the 3-month study window

46.8% of consultations used the AI Consult tool (Jul–Sep 2024)

GPT-4o off-the-shelf LLM, not fine-tuned on Kenyan data

How Clinicians Used It

1. **Select AI Consult**

   Clinician clicks a button in the EMR mid-visit and picks a prompt type (e.g. “Comprehensive Consult”).

2. **LLM generates guidance**

   De-identified visit data (history, vitals, exam) is sent to GPT-4o, which returns differentials, tests, and treatment suggestions.

3. **Clinician decides**

   Clinician reviews the output and freely accepts, edits, or ignores it — full clinical authority stays with the clinician.

Key point: this evaluates the tool's historical outputs in routine use, not a new intervention or a controlled trial.

Study Design at a Glance

Design	Retrospective observational study of routinely collected EMR data; no control arm, no randomization
Sample	1,469 encounters (age-stratified random sample from 36,670 AI-assisted visits), Jul–Sep 2024
Reviewers	30 independent Kenyan physicians, trained & calibrated against a consensus reference standard
What was scored	(1) quality of clinician's initial documentation, (2) LLM diagnostic reasoning, (3) LLM clinical/management reasoning, (4) safety (harmful vs. beneficial outputs, and whether adopted)

Note: *clinicians were never blinded and always retained full authority. This is a study of real clinical behavior around the tool, not of the tool in isolation.*

Results: Following the Encounter Through

1,469 encounters, reviewed by an independent physician panel

1. Baseline documentation

60% / 17%

rated acceptable / high quality

37%

of initial notes had a safety concern (most commonly inappropriate medication, 49%)

4%

totally inadequate documentation

2. LLM output quality

83% / 14%

responses fully / mostly addressed the clinical issue

99%

of management advice aligned with local guidelines

3.4%

contained hallucinations — mostly misexpanded acronyms or drug names

3. LLM safety concerns

7.8%

of responses had actively harmful recommendations (2.5% major, 5.3% minor)

46%

of harmful outputs were inappropriate medication suggestions

8.0%

of encounters had initial-note risk fully/partially mitigated by the LLM

4. Clinician adoption & outcome

62%

of encounters: no documentation change after AI Consult

58% vs 22%

harmful advice acted on far more often than beneficial advice

0

adverse patient outcomes confirmed on follow-up review

The asymmetry: of the encounters where clinicians acted on AI advice, beneficial recommendations were followed only 22% of the time they were offered. Harmful ones, 58% of the time. Total inference cost for all 1,469 encounters: $7.81 (~0.5¢/encounter).

The Headline Tension — For Discussion

83%

of LLM responses fully addressed the clinical issues in the note

7.8%

of responses contained actively harmful recommendations

58%

of harmful AI outputs were followed by clinicians — vs. only 22% of beneficial ones

Beneficial advice was more available, but harmful advice was more likely to be acted on. Why might that asymmetry exist, and what would fix it?

Other Discussion Questions

1. This paper doesn't address deskilling and doesn't have the longitudinal nature to observe it. Do we think that is a risk here?
2. In this study clinicians maintained 'full authority.' However, the AI output still got embedded into the documentation and acted in a few instances. Who's accountable if something goes wrong?
3. One major aspect that they highlight is non-generalizability to the larger Kenyan health system. What concerns may be raised in a lower-resourced setting in acting on AI advice?

IGHS Video Tutorials

http://tiny.ucsf.edu/a5P3ZH

https://tiny.ucsf.edu/a5P3Z

Further IGHS Support: Chat Champions

- Andrea DeLuca
- CC Lipp
- Melissa Paschuck
- Josh Richardson
- Kelly Taylor
- Paul Wesson

On Teams at: IGHS Chat Champions

http://tiny.ucsf.edu/vZmhYL

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